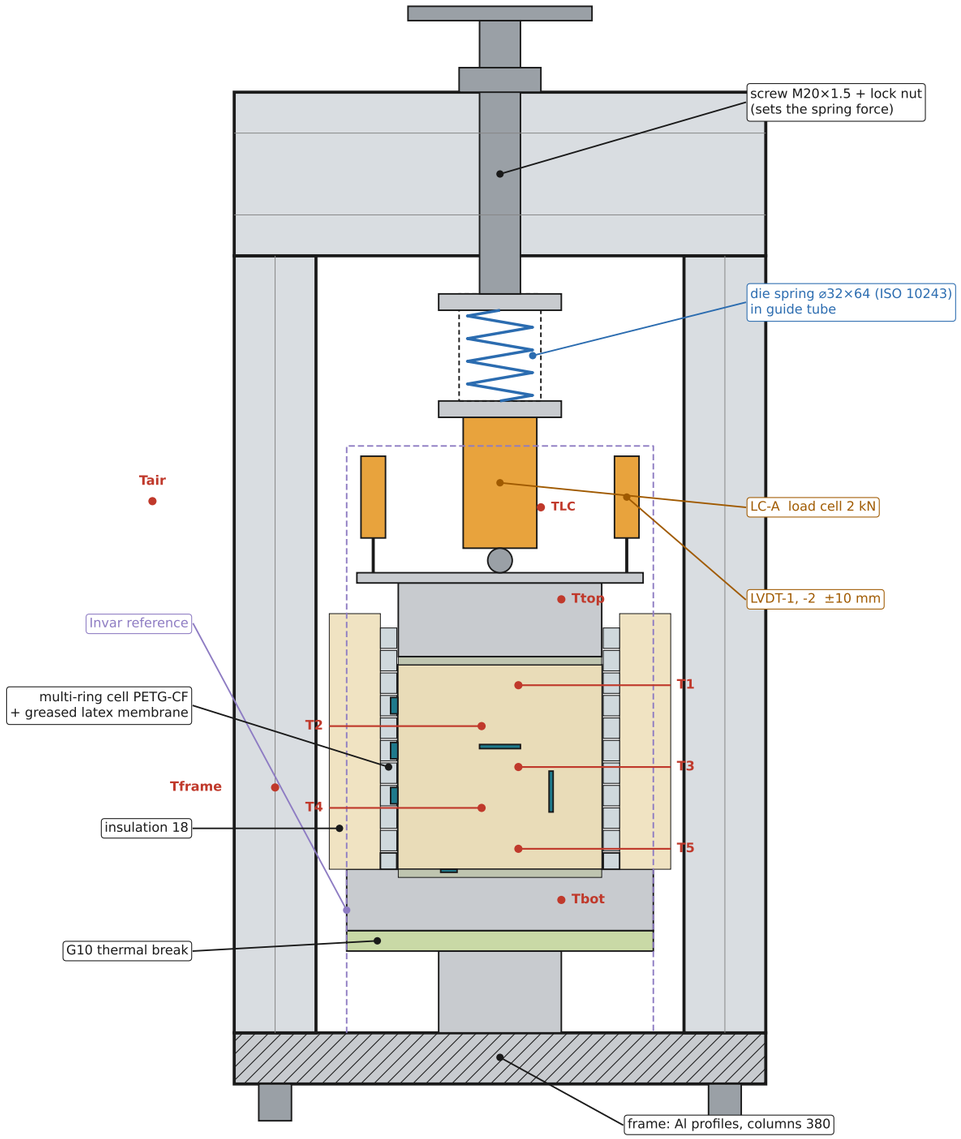


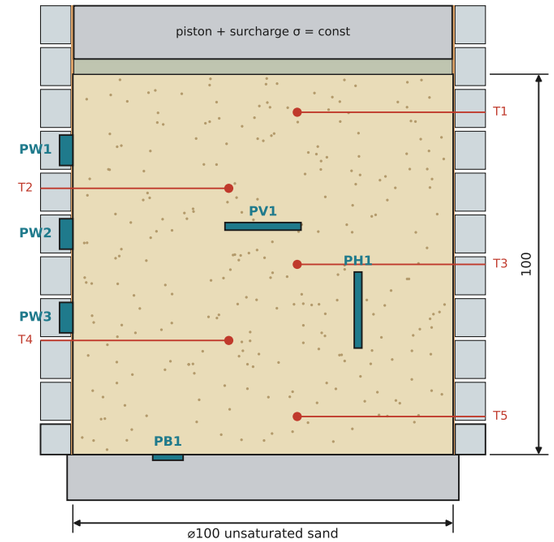
Sensors (tag → model → CR1000)

tag	measures	recommended model	CR1000
LC-A	spring force	Interface SSM 2 kN	DIFF1, BrFull, VX1
PV1 / PH1	vert. / radial stress	TML PDA-PA 200 kPa on ⌀20×2 disc	MUX-1 (4×16), BrFull, VX3 2 V
PB1	vertical stress at base	TML PDA-PA 200 kPa (flush)	MUX-1 (4×16), BrFull, VX3 2 V
PW1…PW3	lateral stress on wall	TML PDA-PA 200 kPa (flush)	MUX-1 (4×16), BrFull, VX3 2 V
LVDT-1,2	check: no movement	RDP DCTH400	SE7–10, VoltSE ±5 V
T1…T5	soil, r 9 mm	Campbell 109SS	MUX-2 → SE13, VX2
Ttop, Tbot	piston, pedestal	Campbell 109SS	(same)
TLC, Tframe, Tair	load cell, frame, air	Campbell 109SS / 107	MUX-2 → SE13, VX2

spring	σ kPa	F N	compr. mm	Δσ/0.1 mm
light, 20 N/mm	25–50	196–393	9.8–19.6	0.25
heavy, 80 N/mm	100–200	785–1571	9.8–19.6	1.02

How it works
Turn the screw until LC-A shows the target force (50 kPa = 393 N), lock the nut.
The spring is soft against the sand: 0.1 mm movement changes σ by 0.25 kPa,
the 0.18 mm shrinkage of the Al frame in the cold by < 0.5 kPa (light spring).
Embedded discs (PV1, PH1), the base sensor (PB1) and the wall sensors (PW)
measure the stress rise caused by ice in the pores of the unsaturated sand.

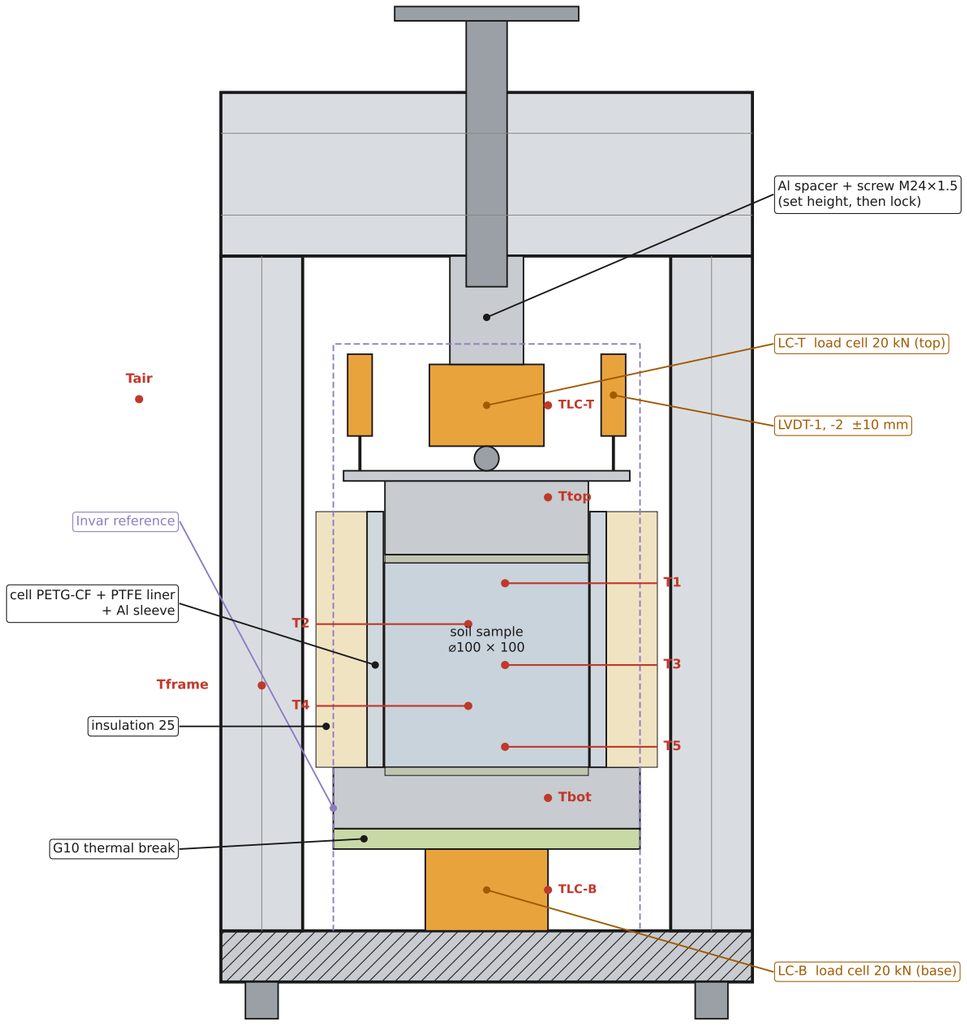
DETAIL – sand, pressure sensors, thermistors ≈ 1 : 1.6



schematic; orange = force / displacement, blue = pressure, red = temperature, dark blue = spring; cells revolved into section

Variant A – unsaturated sand, spring surcharge, internal pressures

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	≈ 1 : 3	drawing no.	**E7-V01**	sheet 1 / 6



Sensors (tag → model → CR1000)

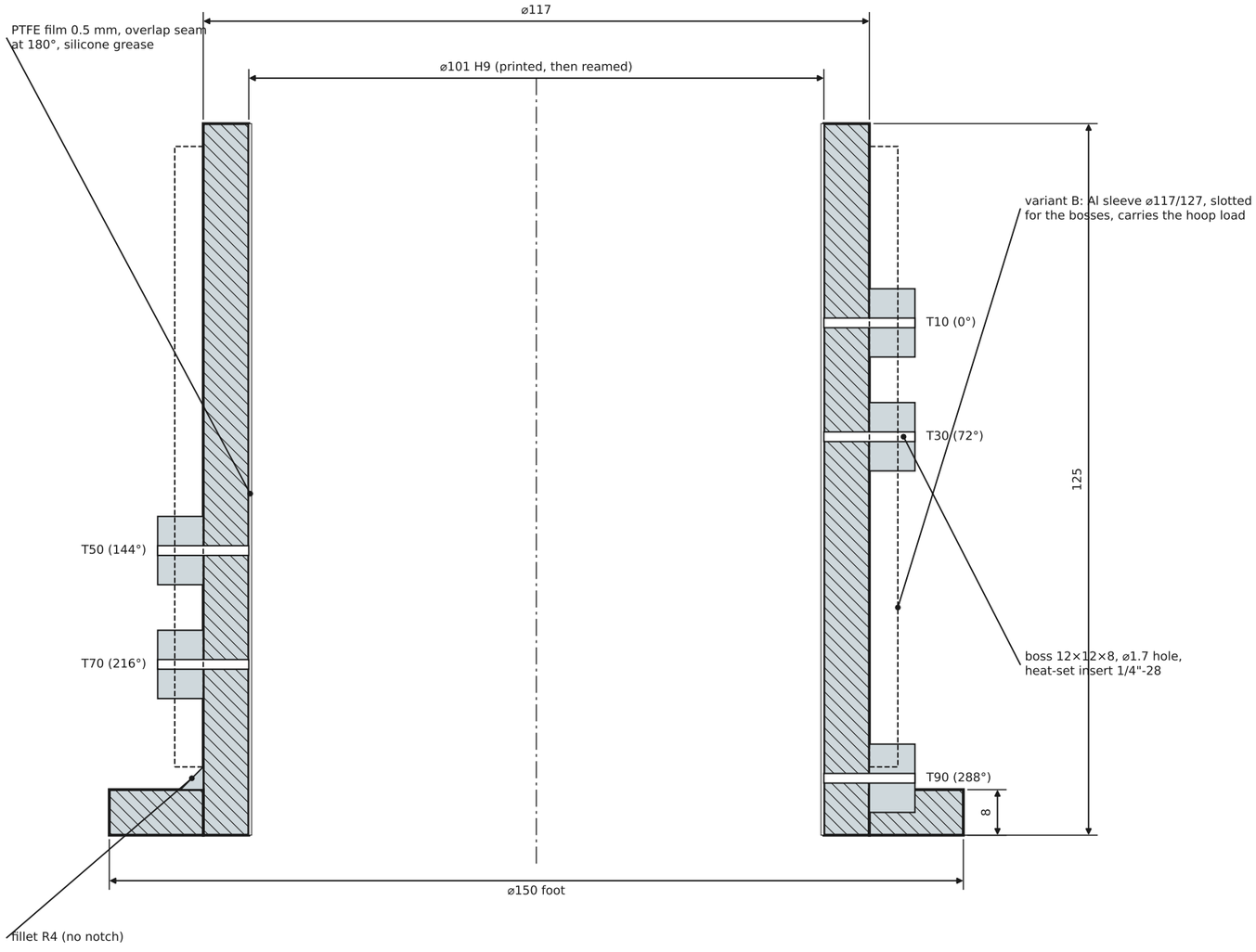
tag	measures	recommended model	CR1000
LC-T	heave force, top	HBM C9C/20kN	DIFF2, BrFull, VX1
LC-B	force at base	HBM C9C/20kN	DIFF3, BrFull, VX1
LVDT-1,2	residual movement	RDP DCTH400	SE7–10, VoltSE ±5 V
T1…T5	soil, r 9 mm	Campbell 109SS	MUX-2 → SE13, VX2
Ttop, Tbot	piston, pedestal	Campbell 109SS	(same)
TLC-T/B, Tframe, Tair	load cells, frame, air	Campbell 109SS / 107	MUX-2 → SE13, VX2

How it works
The screw holds the piston at constant height; the soil pushes against it while freezing. LC-T measures that force, LC-B the force arriving at the base – the difference is the friction on the cell wall. The LVDTs show how much the frame still gives (report the stiffness with the pressure).
Run a blank test with an Al dummy to remove the effect of the frame shrinking in the cold (vypocty.md §3).

schematic, proportions to scale; sensors in orange (force, displacement) and red (temperature)

Variant B – constant height (simplified)

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**≈ 1 : 3**	drawing no.	**E7-V02**	sheet 2 / 6



Material check – PETG-CF

load case	stress	SF short	SF long
hoop stress, A 200 kPa	1.3 MPa	> 30	> 9
hoop stress, B 1 MPa	6.3 MPa	6–9	1.9–2.6
hoop stress, B 2 MPa	12.6 MPa	3–4	1.0–1.3
hoop, B 2 MPa + Al sleeve 5 mm	≈ 1 MPa*	> 40	> 12
axial (Z), friction μ 0.5, 1 MPa	5.8 MPa	2.6–5	< 1.6
axial (Z), PTFE μ 0.05, 1 MPa	0.6 MPa	> 25	> 7

Assumed strength (data-sheet range, verify): XY 40–55 MPa, Z 15–30 MPa, E 3–6 GPa; long-term limit ≈ 30 % (creep, layer bond). Details: vypocty.md §4.

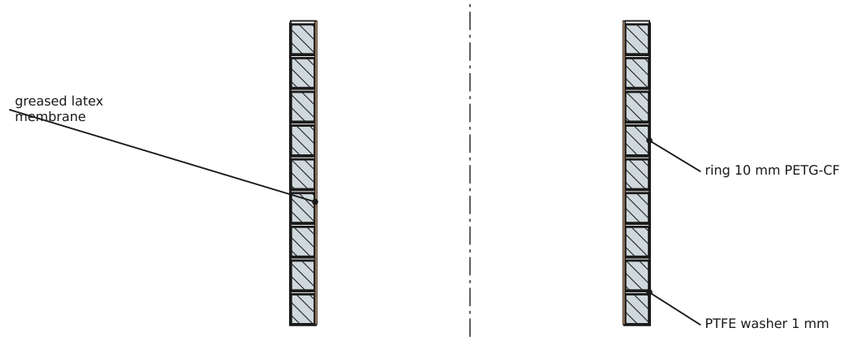
Verdict: suitable for variant A; for B (design 2 MPa) only with the Al sleeve shown dashed (*it carries ≈ 90 % of the hoop load: 70 GPa×5 mm vs. ≈ 4 GPa×8 mm), the PETG-CF then acts as a liner.
Conditions for both:
- the wall is ≥ 8 mm and printed with concentric perimeters only (100 % infill, no zig-zag), so the hoop stress runs along the filament;
- friction is kept low (PTFE liner) and the cell is not clamped at the foot, so friction pushes the wall instead of pulling the layers apart;
- the inside is sealed (thin epoxy coat) so water cannot enter print voids and split layers when it freezes;
- there are no printed threads or sharp notches (CF lowers toughness at −10 °C) – heat-set inserts and fillets instead.

Variant A: a multi-ring cell (view F) is the literature standard against wall friction – printed rings 10 mm + PTFE washers + greased latex.

Print settings (starting point)
hardened nozzle 0.6 mm, layer 0.2 mm, 250–260 °C, bed 80 °C, slow outer walls, filament dried 4 h at 65 °C, enclosure, part cooling ≤ 30 %.

Qualification before use
1 ring burst test: printed ring 20 mm high, internal water pressure to failure (target ≥ 3× design pressure);
2 10 freeze–thaw cycles −10/+10 °C of water-saturated coupons, then tensile test XY and Z vs. dry control;
3 creep: ring under 1 MPa for 7 days at +10 °C, hoop strain gauge.

VIEW F – multi-ring cell (variant A) 1 : 2

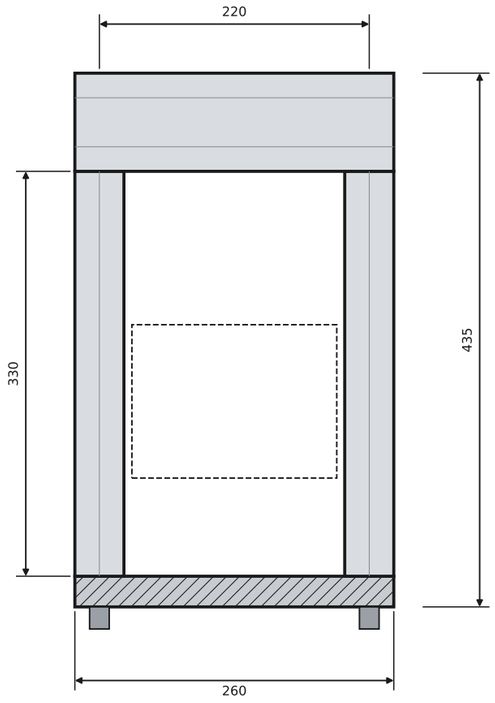


print upright; hoop direction = layer direction; foot rests on the pedestal, not bolted

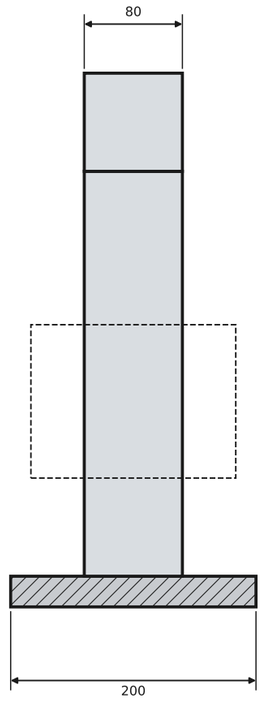
Sample cell – PETG-CF, 3D printed

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 1**	drawing no.	**E7-V03**	sheet 3 / 6

FRONT VIEW 1 : 5



SIDE VIEW 1 : 5



Bill of materials

#	part	size	qty
1	base plate Al 6082 T6	260 × 200 × 25	1
2	profile 40×80 (column)	l = 330 (B) / 380 (A)	2
3	profile 80×80 (cross-beam)	l = 260	1
4	core bolt M12 into tapped profile end		8
5	socket screw M10 base → column end		4
6	levelling foot M12, steel		3
7	Invar rod ⌀10 + Invar/Al bridge (LVDT)	l = 300	2 + 1
8	nut M20×1.5 (A) / M24×1.5 (B)	in cross-beam	1
9	end caps, T-slot covers		–

Profile values used in vypocty.md §2 are typical catalogue values (series 8): 40×80 A ≈ 1050 mm², 80×80 I ≈ 1.4·10⁶ mm⁴ – check against the chosen supplier.

Stiffness at 15.7 kN (2 MPa): cross-beam 0.036 mm, columns 0.035 mm, load cell ≈ 0.05 mm, joints 0.02–0.15 mm. Joints dominate → core bolts into tapped profile ends, torque-controlled, no angle brackets; optionally 2 steel tie rods M12 next to the cell.

Fit in the chest freezer: frame 260 × 200, height 435 mm incl. base; variant B; variant A has 380 mm columns (spring); two rigs side by side on the tray – layout and heights on E7-V06.

one frame for both variants; core-bolted joints, no angle brackets in the load path

Frame – aluminium profiles

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 5**	drawing no.	**E7-V04**	sheet 4 / 6

Options per measurement (PICK = recommended)

sensor	maker / model	key specs	verdict
LC 2 kN	Interface SSM 2 kN	comp. −15…+65 °C, op. −55…+90 °C, IP67	PICK
LC 2 kN	HBM S9M/2kN	class 0.02, 2 mV/V, 6-wire, IP68, nom. −10…+70 °C	alternative
LC 2 kN	HBM U2B 2 kN	class 0.1, 2 mV/V, IP67, temp. [U]	alternative
LC 2 kN	ME KD80s 2 kN	2 mV/V, 0.2 mm at FS, −10…+70 °C	alternative
LC 2 kN	Lorenz K-25	1–2 mV/V, IP67, ext. range on request	ask quote
LC 2 kN	AML DBBSM	comp. only 0…+70 °C, IP51	not suitable
LC 20 kN	HBM C9C/20kN	⌀26×13 mm, 1 mV/V, nom. −10…+70, op. −30…+85 °C	PICK
LC 20 kN	Burster 8402 20 kN	1.5 mV/V, ≤ 50 µm deflection, op. −30…+100 °C, IP54	alternative
LC 20 kN	HBM C10 25 kN	class 0.02–0.05, height 36 mm [U]	alternative
LC 20 kN	Futek LTH500 (≈22 kN)	donut, 0.05 mm, op. −50…+93 °C	alternative
LC 20 kN	Interface LBM	comp. only +21…+77 °C	not suitable
LVDT ±10	RDP DCTH400	±5 V / 0–10 V out, −40…+80 °C, ±0.25 % FS	PICK
LVDT ±10	Micro-Epsilon DTA-10D + MSC7401	0–5 V / 4–20 mA, −20…+80 °C	alternative
LVDT ±10	Solartron AX/10/S + OD5	±5 V out, needs 230 V conditioner	alternative
LVDT ±10	HBM WA-L 20 mm	passive, needs carrier amplifier	alternative
LVDT ±10	TE DC-EC 500	operating only 0…+70 °C	not suitable
p cell	TML PDA-PA 200 kPa	⌀6.5×1 mm, 0.8 mV/V, comp. −10…+60, op. −20…+70 °C	PICK
p cell	Kyowa BEC-A 200/500 kPa	⌀30 mm, for model tests; t, temp. [U]	one cell only
p cell	TML KDE/KDF-PA	⌀50 mm – too large for ⌀100 sample	not suitable
p wall	TE/MEAS EPL	miniature, comp. −40…+90 °C, ⌀ [U]	alternative
p cell	Tekscan FlexiForce A201	thin film, drift < 5 %/log t	comparison only
T probe	Campbell 109SS	⌀1.6 mm, ±0.38 °C at 0 °C, direct on CR1000	PICK
T probe	Campbell 107	⌀7.6 mm, ±0.2 °C (0–50 °C)	air / frame
T probe	TE/YSI 44031 bead	±0.1 °C, own half bridge needed	for 0.01 °C (E6 chain)
T probe	Pt100 1/10 DIN + 4WPB100	2 DIFF + VX per sensor	too many channels

CR1000 channel plan

terminal	connected to	measurement
DIFF1–3	LC-A, LC-T, LC-B	BrFull, reversal; VX1 2500 mV
SE7–SE10	LVDT-A1, A2, B1, B2	VoltSE ±5000 mV, averaging; own ±15 V supply
DIFF6 (SE11/12)	MUX-1 AM16/32B, 4×16	6 pressure bridges (PV1, PH1, PB1, PW1–3), BrFull
VX3	MUX-1 excitation	2000 mV (TML: < 2 V recommended)
C1, C2	MUX-1 reset, clock	
SE13	MUX-2 AM16/32B, 2×32	≈ 24 thermistors (Therm109), VX2
C3, C4	MUX-2 reset, clock	
12V, G	both multiplexers	
SE14–16	spare	

Channel count (2 rigs): 3 load cells, 4 LVDTs, 6 pressure bridges, ≈ 24 thermistors.
16 SE / 8 DIFF on the CR1000 → two AM16/32B: bridges (4×16), thermistors (2×32).

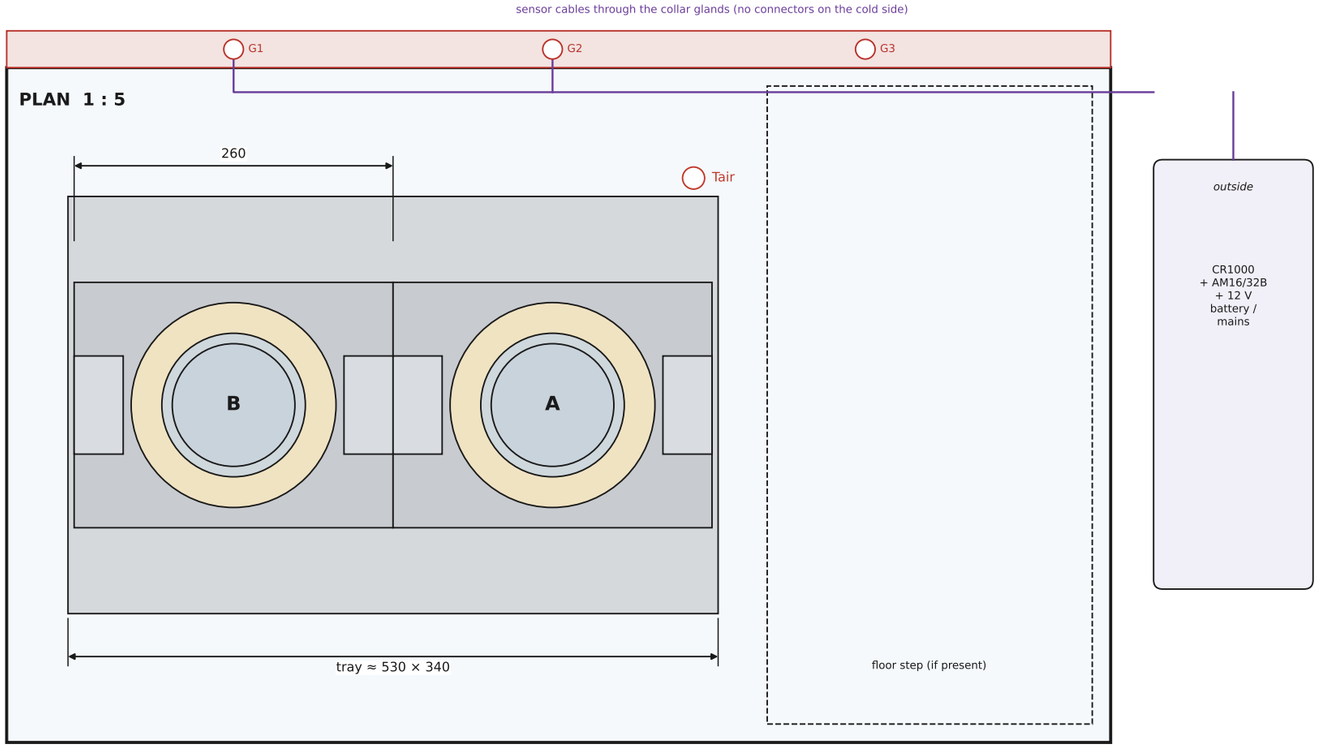
Resolution with the CR1000 (13-bit ADC)
- LC-A 2 mV/V, 2.5 V: 5 mV FS → ≈ 0.3 N ≈ 0.04 kPa
- LC-T/B 1 mV/V: 2.5 mV FS → ≈ 2.6 N ≈ 0.3 kPa
- LVDT ±5 V over ±10 mm: ≈ 1.3 µm (0.7 µm with reversal); absolute linearity ±0.25 % FS ≈ ±50 µm → calibrate with a micrometer
- 109SS: ±0.38 °C at 0 °C; ice-point offset check improves it

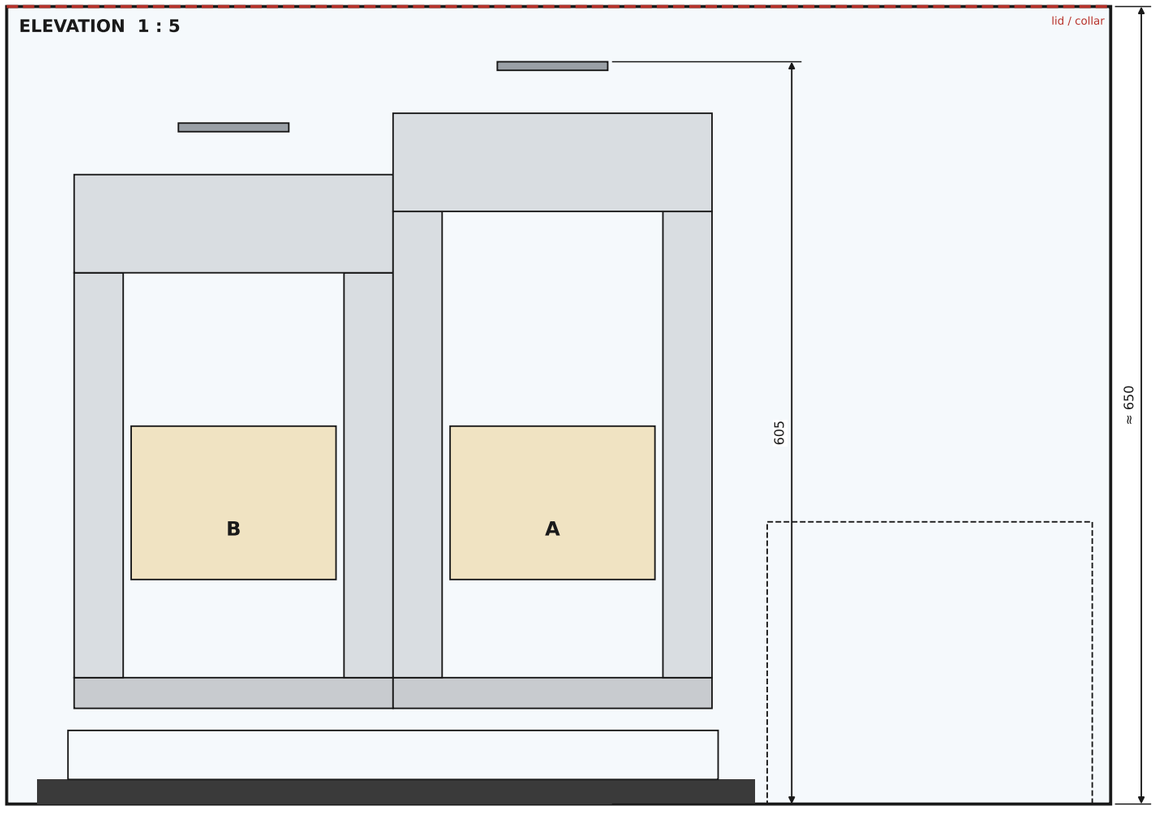
Notes
- CR1000 is discontinued (successor CR1000X / CR1000Xe, 24-bit); the existing one is fine for these resolutions.
- Logger outside the freezer (CR1000 −25…+50 °C).
- No load cell found with compensation to −20/−30 °C as standard: Interface SSM −15 °C, HBM −10 °C (op. −30 °C) – log TLC and correct; Interface/Lorenz offer extended ranges on request.
- 4WFBS modules are for quarter bridges – not needed here.
- Pressure cells: calibrate in the same sand under dead weight (25–200 kPa) at +10 and −10 °C; correct zero(T) with the neighbouring thermistor.

both rigs on one CR1000 + AM16/32B; models from manufacturer data (see README §7)

Sensor selection and CR1000 wiring

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	–	drawing no.	**E7-V05**	sheet 5 / 6





Fit check

item	size	limit	status
base plates 2 × 260 × 200	520 × 200	tray ≈ 530 × 340	OK (tight)
rig A (spring)	nothing outside the frame	–	OK
highest point	605 mm	inside ≈ 650	OK, measure
logger	outside, 0…+40 °C	CR1000 −25…+50 °C	keep warm & dry

Notes
- Both rigs share the freezer program (Set A) and the CR1000; each rig has its own Tair sensor at sample mid-height and its own load cell temperatures.
- Both rigs are set from above with a screw and handwheel; the free end of the freezer stays empty (room for the cable bundle or a third rig).
- Run the blank test with Al dummies in both rigs before the first soil run.
- Dimensions marked ≈ are estimates from the photos (E6) – please measure.

same freezer as E6 (inside ≈ 900 × 550 × 650, to be measured); logger outside

Both rigs in the chest freezer + CR1000

status	DRAFT	drawn	Claude Code (draft)	checked: —
units	mm	project	E7 – freeze–thaw loading tests	date 2026-10-20
scale	**1 : 5**	drawing no.	**E7-V06**	sheet 6 / 6